Formula and Quantitative Estimation Handbook

From first principles to expert practice

Built from the manuscript sources · 0 chapter(s) in this edition

Appendix C — Formula and Quantitative Estimation Handbook

Chapter 1 — What Is System Design?
Chapter 2 — How to Think Like a System Designer
Chapter 3 — The System Design Learning Method
Chapter 4 — Quantitative Foundations for System Design
Chapter 5 — Computer Architecture
Chapter 6 — Operating Systems
Chapter 7 — Networking Fundamentals
Chapter 8 — Web Application Fundamentals
Chapter 9 — Real-Time Communication
Chapter 10 — Security Fundamentals for System Designers
Chapter 11 — Relational Databases
Chapter 12 — NoSQL Databases
Chapter 13 — Storage Engine Internals
Chapter 14 — Database Scaling: Replication, Partitioning and Sharding
Chapter 15 — Object, Block and File Storage
Chapter 16 — Search Systems
Chapter 17 — Data Structures and Algorithms for Scale

Appendix C — Formula and Quantitative Estimation Handbook

> Generated by `build/build.py` from every `formula` callout in the manuscript, grouped by chapter. Worked examples remain in the chapters; follow the section reference.

Chapter 1 — What Is System Design?

From §1.3.2

FORMULA

Average and peak request rate

$$\mathrm{RPS}_{\mathrm{avg}} = \frac{\mathrm{DAU} \times r}{86,400} \qquad \mathrm{RPS}_{\mathrm{peak}} = k \times \mathrm{RPS}_{\mathrm{avg}}$$

Symbol	Meaning	Units
$\mathrm{RPS}_{\mathrm{avg}}$	Average request rate over a day	requests/second
DAU	Daily active users	users
r	Requests (of the type being estimated) per user per day	requests/user/day
$86,400$	Seconds in a day	seconds/day
k	Peak factor: ratio of the busiest period's rate to the daily average (dimensionless; 2–5 is a common assumption for consumer apps)	—
$\mathrm{RPS}_{\mathrm{peak}}$	Request rate during the busiest period	requests/second

From §1.7.2

FORMULA

Availability and downtime budget

$$A = \frac{\text{MTBF}}{\text{MTBF}+\text{MTTR}} \qquad D = (1 - A) \times T$$

Symbol	Meaning	Units
A	Availability: fraction of time the service is up (dimensionless, 0–1)	—
MTBF	Mean time between failures	hours
MTTR	Mean time to repair (restore service)	hours
D	Allowed downtime in the period	minutes (or hours)
T	Length of the period	minutes (or hours)

From §1.7.2

FORMULA

Availability of components in series (independent failures)

$$A_{\text{series}} = \prod_{i=1}^{n} A_i$$

Symbol	Meaning	Units
A_{series}	Availability of the whole request path	—
A_i	Availability of component i	—
n	Number of components the request depends on	count

From §1.8.1

FORMULA

Weighted decision score

$$S_j = \sum_{i=1}^{m} w_i \, s_{ij}, \qquad \sum_{i=1}^{m} w_i = 1$$

Symbol	Meaning	Units
S_j	Total score of option j	points (same scale as s_{ij})
w_i	Weight of criterion i (importance; sums to 1)	—
s_{ij}	Score of option j on criterion i (here 1 = poor, 5 = excellent)	points
m	Number of criteria	count

Chapter 2 — How to Think Like a System Designer

From §2.5.1

FORMULA

Allowed downtime for an availability target

$$T_{down} = (1 - A) \times T_{period}$$

Symbol	Meaning	Units
T_{down}	Maximum downtime allowed in the period	minutes
A	Availability target, as a fraction	dimensionless (e.g. 0.9995)
T_{period}	Length of the measurement period	minutes

From §2.6.1

FORMULA

Service demand, the utilization law and the bottleneck bound

$$D_i = V_i \times S_i \qquad U_i = X \times D_i \qquad X_{max} = \frac{c_i}{D_{max}}$$

Symbol	Meaning	Units
D_i	Service demand: total busy time one request places on resource i	seconds per request
V_i	Visits: how many times one request uses resource i	visits per request
S_i	Service time per visit to resource i	seconds per visit
U_i	Utilization of resource i, per unit of capacity	fraction (0–1)
X	System throughput	requests per second
c_i	Number of parallel units of resource i (e.g. cores)	units
X_{max}	Maximum throughput, set by the resource with the largest demand per unit	requests per second

The utilization law for a resource with c_i parallel units is $U_i = X \times D_i / c_i$. The bottleneck is the resource with the largest D_i / c_i.

From §2.6.1

FORMULA

Little’s Law (concurrency)

$$L = \lambda \times W$$

Symbol	Meaning	Units
L	Average number of requests in the system (or holding a resource) at once	requests
λ	Average arrival rate	requests per second
W	Average time each request spends in the system (or holds the resource)	seconds

Little’s Law holds for any stable system, regardless of the arrival pattern [Little 1961]. Full treatment in §4.11.

From §2.8.1

FORMULA

Weighted decision score

$$S_j = \sum_{i=1}^{n} w_i \times s_{ij}, \qquad \sum_{i=1}^{n} w_i = 1$$

Symbol	Meaning	Units
S_j	Total weighted score of option j	score points (same scale as s_{ij})
w_i	Weight of criterion i	fraction (0–1)
s_{ij}	Score of option j on criterion i	score points, e.g. 1–5
n	Number of criteria	count

From §2.9.1

FORMULA

Availability of components in series

$$A_{series} = \prod_{i=1}^{n} A_i$$

Symbol	Meaning	Units
A_{series}	Availability of a request path that needs *all* n components	fraction (0–1)
A_i	Availability of component i	fraction (0–1)
n	Number of components on the synchronous path	count

This assumes independent failures. Parallel (redundant) components and correlated failures are treated in §4.14.

From §2.11.1

FORMULA

Average and peak request rate

$$R_{avg} = \frac{U \times a}{86,400} \qquad R_{peak} = \frac{U \times a \times h}{3,600} \times b$$

Symbol	Meaning	Units
R_{avg}	Average request rate over a day	requests/s
R_{peak}	Design peak request rate	requests/s
U	Daily active users	users
a	Actions of this type per user per day	requests/user/day
h	Share of daily volume in the busiest hour	fraction
b	Burst factor above the busiest-hour average	dimensionless
86,400 / 3,600	Seconds per day / per hour	s

From §2.11.1

FORMULA

Storage and node count

$$S = n_{day} \times s \times T \times k \qquad N = \lceil \frac{R_{peak}}{c \times u} \rceil + N_{spare}$$

Symbol	Meaning	Units
S	Total storage needed	bytes
n_{day}	Records created per day	records/day
s	Size per record, including indexes and overhead	bytes
T	Retention period	days
k	Number of stored copies (replication)	dimensionless
N	Nodes needed	nodes
c	Throughput one node sustains	requests/s
u	Target utilization, leaving head-room for queueing (§4.10)	fraction
N_{spare}	Extra nodes so a failure or deploy does not overload the rest	nodes

Chapter 3 — The System Design Learning Method

From §3.3.2

FORMULA

Fraction of keys remapped when growing from N to $N+1$ nodes

$$f_{\text{mod}} = 1 - \frac{1}{N+1} = \frac{N}{N+1} \qquad f_{\text{ring}} \approx \frac{1}{N+1}$$

Symbol	Meaning	Units
N	Number of nodes before the change	nodes
f_{mod}	Expected fraction of keys whose owner changes under `hash mod N` (uniform hashing)	dimensionless (0–1)
f_{ring}	Expected fraction of keys whose owner changes under consistent hashing	dimensionless (0–1)

From §3.9.1

FORMULA

Gap priority

$$P_i = (T_i - C_i) \times W_i$$

Symbol	Meaning	Units
P_i	Priority score for part i	points (0–12)
T_i	Target proficiency for part i	0–4 scale
C_i	Current self-rated proficiency (use 0 if $C_i > T_i$ would make P_i negative)	0–4 scale
W_i	Relevance weight: 1 = nice to have, 2 = useful, 3 = essential for role	dimensionless (1–3)

Chapter 4 — Quantitative Foundations for System Design

From §4.3.1

FORMULA

Laws of exponents

$$b^m \times b^n = b^{m+n} \qquad \frac{b^m}{b^n} = b^{m-n} \qquad (b^m)^n = b^{mn}$$

Symbol	Meaning	Units
b	Base (2 for binary quantities, 10 for decimal)	dimensionless
m, n	Exponents	dimensionless

From §4.4.2

FORMULA

Logarithm rules and change of base

$$\log_b (xy) = \log_b x + \log_b y \qquad \log_b (x^k) = k \log_b x \qquad \log_b x = \frac{\ln x}{\ln b} = \frac{\log_{10} x}{\log_{10} b}$$

Symbol	Meaning	Units
b	Base of the logarithm ($b > 0$, $b \neq 1$)	dimensionless
x, y	Positive numbers	dimensionless
k	Any real exponent	dimensionless
$\ln$	Natural logarithm, base $e \approx 2.71828$	—

From §4.4.3

FORMULA

Height of a B-tree

$$h = \lceil \log_f n \rceil = \lceil \frac{\log n}{\log f} \rceil$$

Symbol	Meaning	Units
h	Number of levels from root to leaf (page reads per lookup if nothing is cached)	levels
n	Number of keys indexed	keys
f	Fan-out: child pointers per internal node	children/node
$\lceil \cdot \rceil$	Ceiling: round up to the next integer	—

From §4.5.1

FORMULA

Converting link speed to byte throughput

$$R_{\text{B/s}} = \frac{R_{\text{bps}}}{8} \qquad t_{\text{transfer}} = \frac{8 \times S}{R_{\text{bps}}}$$

Symbol	Meaning	Units
R_{bps}	Link rate	bits/second
$R_{\text{B/s}}$	The same rate expressed in bytes	bytes/second
S	Amount of data to transfer	bytes
t_{transfer}	Ideal transfer time, ignoring protocol overhead and congestion	seconds

From §4.5.3

FORMULA

Average and peak rate from a daily volume

$$\bar{\lambda} = \frac{V_{\text{day}}}{86,400} \qquad \lambda_{\text{peak}} = k \times \bar{\lambda}$$

Symbol	Meaning	Units
V_{day}	Events (requests, writes, messages) per day	events/day
$\bar{\lambda}$	Average rate	events/second
k	Peak-to-average ratio, measured or assumed (often 2–10 for consumer traffic)	dimensionless
λ_{peak}	Rate the system must be sized for	events/second

From §4.6.1

FORMULA

Probability that at least one of N independent components fails

$$P(\text{at least one fails}) = 1 - (1 - p)^N$$

Symbol	Meaning	Units
p	Probability that a single component fails during the period of interest	dimensionless (0–1)
N	Number of independent components	count
$(1 - p)^N$	Probability that *none* fails (complement, multiplied N times by independence)	dimensionless

From §4.6.2

FORMULA

Union (inclusion-exclusion) and intersection

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

Symbol	Meaning	Units
$P(A \cup B)$	Probability that A or B (or both) happens	dimensionless
$P(A \cap B)$	Probability that both happen	dimensionless
$P(B \mid A)$	Conditional probability of B given that A happened; equals $P(B)$ when independent	dimensionless

From §4.6.3

FORMULA

Expected value and linearity of expectation

$$E[X] = \sum_{i} x_i \, P(X = x_i) \qquad E[X_1 + X_2 + \cdots + X_N] = E[X_1] + \cdots + E[X_N]$$

Symbol	Meaning	Units
X	A random quantity (failures, latency, cost)	units of the quantity
x_i	A possible value of X	units of the quantity
$P(X = x_i)$	Probability of that value	dimensionless

From §4.6.4

FORMULA

Binomial probability of exactly k failures

$$P(K = k) = \binom{N}{k} p^k (1-p)^{N-k}, \qquad \binom{N}{k} = \frac{N!}{k!\,(N-k)!}, \qquad E[K] = Np$$

Symbol	Meaning	Units
N	Number of independent components (trials)	count
k	Number of failures we ask about	count
p	Per-component failure probability in the period	dimensionless
$\binom{N}{k}$	Number of ways to choose which k components fail ("N choose k")	count
$E[K]$	Expected number of failures	count

From §4.7.1

FORMULA

Mean, variance and standard deviation

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2 \qquad \sigma = \sqrt{\sigma^2}$$

Symbol	Meaning	Units
x_i	The i-th measurement	e.g. ms
n	Number of measurements	count
$\bar{x}$	Mean	same as x (ms)
σ^2	Variance: average squared distance from the mean	ms²
σ	Standard deviation: typical distance from the mean	ms

From §4.7.2

FORMULA

Tail formulas for three skewed distributions

Exponential: $P(T > t) = e^{-\lambda t}$, $E[T] = \frac{1}{\lambda}$ Log-normal: median $= e^{\mu}$, mean $= e^{\mu + s^2/2}$ Pareto: $P(X >$

Symbol	Meaning	Units
λ	Event rate of the exponential distribution	events/second
t	Elapsed time	seconds
μ, s	Mean and standard deviation of $\ln X$ for a log-normal X	ln of the unit of X
x_m	Pareto minimum (scale) value	unit of X (e.g. bytes)
α	Pareto tail index: smaller means heavier tail	dimensionless

From §4.8.2

FORMULA

Nearest-rank percentile

$$r = \lceil \frac{q}{100} \times n \rceil, \qquad \mathrm{p}q = x_{(r)}$$

Symbol	Meaning	Units
q	Desired percentile, $0 < q \le 100$	percent
n	Number of samples	count
r	1-based rank in the sorted list	position
$x_{(r)}$	The r-th smallest sample	e.g. ms

From §4.8.4

FORMULA

Probability that a fan-out request is fast

$$P(\text{all } N \text{ fast}) = p^N \qquad P(\text{at least one slow}) = 1 - p^N \qquad p_{\text{required}} = P_{\text{target}}^{1/N}$$

Symbol	Meaning	Units
p	Probability that one leaf responds within the threshold (e.g. 0.99 at the leaf's p99)	dimensionless
N	Number of parallel sub-requests the root must wait for	count
P_{target}	Desired probability that the whole request is fast	dimensionless
p_{required}	Per-leaf probability needed to meet P_{target}	dimensionless

From §4.9.2

FORMULA

Dot product and cosine similarity

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{d} a_i b_i \qquad \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|}$$

Symbol	Meaning	Units
$\mathbf{a}, \mathbf{b}$	Two vectors of the same dimension	(embedding units; dimensionless)
d	Dimension	count
a_i, b_i	i-th components	dimensionless
$\|\mathbf{a}\|$	Length of $\mathbf{a}$	dimensionless
θ	Angle between the vectors	radians or degrees
$\cos\theta$	Cosine similarity: 1 = same direction, 0 = unrelated (perpendicular), −1 = opposite	dimensionless

From §4.9.3

FORMULA

Cost of matrix multiplication

$$\text{FLOPs} = 2mnk \qquad \text{arithmetic intensity} = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Symbol	Meaning	Units
m, k	Rows and columns of A	count
k, n	Rows and columns of B (inner dimension k must match)	count
FLOPs	Floating-point operations: one multiply and one add per term	operations
bytes moved	Data read from and written to memory	bytes
arithmetic intensity	FLOPs per byte; low values mean memory-bandwidth-bound, high values compute-bound	FLOP/byte

From §4.10.2

FORMULA

M/M/1 results

$$\rho = \frac{\lambda}{\mu} \qquad W = \frac{1}{\mu - \lambda} = \frac{S}{1 - \rho} \qquad W_q = \frac{\rho}{\mu - \lambda} \qquad L = \frac{\rho}{1 - \rho} \qquad L_q = \frac{\rho^2}{1 - \rho}$$

Symbol	Meaning	Units
λ	Average arrival rate	jobs/second
μ	Average service rate of the server	jobs/second
$S = 1/\mu$	Average service time	seconds
ρ	Utilization: fraction of time the server is busy (must be < 1)	dimensionless
W	Average time in system (waiting + service), i.e. response time	seconds
W_q	Average time waiting in the queue before service starts	seconds
L	Average number of jobs in the system	jobs
L_q	Average number of jobs waiting in the queue	jobs

From §4.10.4

FORMULA

M/M/c (Erlang C)

$$a = \frac{\lambda}{\mu}, \quad \rho = \frac{a}{c}, \quad P_{\text{wait}} = \frac{\frac{a^c}{c!\,(1-\rho)}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!\,(1-\rho)}}, \quad W = \frac{P_{\text{wait}}}{c\mu - \lambda} + \frac{1}{\mu}$$

Symbol	Meaning	Units
c	Number of servers sharing one queue	count
μ	Service rate of *each* server	jobs/second
a	Offered load: average number of busy servers	servers (Erlangs)
ρ	Per-server utilization (< 1)	dimensionless
P_{wait}	Probability an arriving job finds all servers busy	dimensionless
W	Mean response time	seconds

From §4.11.1

FORMULA

Little’s Law

$$L = \lambda \times W$$

Symbol	Meaning	Units
L	Average number of items in the system (concurrency, occupancy, in-flight requests)	items
λ	Average arrival rate (equals throughput in a stable system)	items/second
W	Average time an item spends inside the system boundary	seconds

From §4.12.1

FORMULA

Amdahl's Law (fixed problem size)

$$S(N) = \frac{1}{(1-p) + \frac{p}{N}} \qquad \lim_{N \to \infty} S(N) = \frac{1}{1-p}$$

Symbol	Meaning	Units
p	Fraction of the single-processor run time that can be parallelized	dimensionless (0–1)
$1-p$	Serial fraction	dimensionless
N	Number of processors, cores or nodes	count
$S(N)$	Speedup relative to one processor	dimensionless (×)

From §4.12.2

FORMULA

Gustafson's Law (scaled problem size)

$$S_{\text{scaled}}(N) = (1-p) + pN = N - (1-p)(N-1)$$

Symbol	Meaning	Units
p	Fraction of the *parallel run's* time spent in parallel work	dimensionless
$1-p$	Fraction of the parallel run's time that is serial	dimensionless
N	Number of processors	count
S_{scaled}	How much more work is completed in the same time than one processor could do	dimensionless (×)

From §4.13.1

FORMULA

Universal Scalability Law

$$C(N) = \frac{N}{1 + \alpha(N - 1) + \beta N(N - 1)} \qquad N^* = \frac{\sqrt{1 - \alpha}}{\beta}$$

Symbol	Meaning	Units
N	Number of nodes, cores, threads or concurrent users	count
$C(N)$	Relative capacity: throughput at N divided by throughput at 1	dimensionless (×)
α	Contention coefficient: serialization on shared resources (queues, locks)	dimensionless (0–1)
β	Coherency coefficient: pairwise coordination cost (cache coherence, consensus, data exchange)	dimensionless (≥ 0)
N^*	Concurrency at which throughput peaks	count

From §4.14.1

FORMULA

Downtime budget

$$D = (1 - A) \times T$$

Symbol	Meaning	Units
A	Availability target	dimensionless
T	Length of the period	minutes (or any time unit)
D	Allowed downtime in the period (the error budget, in time)	same unit as T

From §4.14.2

FORMULA

Series, parallel and k-of-n availability (independent failures)

$$A_{\text{series}} = \prod_{i=1}^{n} A_i \qquad A_{\text{parallel}} = 1 - \prod_{i=1}^{n} (1 - A_i) \qquad A_{k\text{-of-}n} = \sum_{j=k}^{n} \binom{n}{j} A^j (1 - A)^{n-j}$$

Symbol	Meaning	Units
A_i	Availability of component i	dimensionless
$\prod$	Product over all components	—
n	Number of components	count
k	Minimum number of identical components that must be up	count
A	Availability of each identical component (k-of-n case)	dimensionless

From §4.15.1

FORMULA

Reliability and availability from MTBF and MTTR

$$\lambda = \frac{1}{\text{MTBF}} \qquad R(t) = e^{-\lambda t} \qquad A = \frac{\text{MTBF}}{\text{MTBF+MTTR}} \qquad \text{AFR} = 1 - e^{-8760/\text{MTTF}}$$

Symbol	Meaning	Units
MTBF	Mean operating time between failures	hours
MTTR	Mean time to repair (detect + diagnose + fix + restore)	hours
MTTF	Mean time to failure of a non-repairable part	hours
λ	Constant failure rate	failures/hour
t	Mission time: the period we need it to survive	hours
$R(t)$	Probability of no failure during t	dimensionless
A	Steady-state availability	dimensionless
AFR	Annualized failure rate: probability a part fails within a year (8,760 h)	dimensionless (often %)

From §4.16.1

FORMULA

Approximate annual probability of losing an n-way replicated item

$$P_{\text{loss/yr}} \approx n!\ \lambda^{n}\ T^{n-1} \times H \qquad \text{valid when } \lambda T \ll 1$$

Symbol	Meaning	Units
n	Number of replicas	count
λ	Per-replica failure rate (AFR / 8,760 for small AFR)	failures/hour
T	Mean time to detect and re-replicate a lost copy	hours
H	Hours per year (8,760)	hours/year
$P_{\text{loss/yr}}$	Annual probability that all n replicas are lost	dimensionless

From §4.18.4

FORMULA

Standard error of a sample mean

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

Symbol	Meaning	Units
σ	Standard deviation of the quantity being measured	units of the quantity
n	Number of independent samples	count
SE	Typical error of the estimated mean	units of the quantity

From §4.18.5

FORMULA

Compute-bound token throughput

$$X_{\text{tokens}} = \frac{\text{MFU} \times F_{\text{peak}}}{2P}$$

Symbol	Meaning	Units
F_{peak}	Peak accelerator throughput at the precision used	FLOP/s
MFU	Model FLOPs utilization	dimensionless (0–1)
P	Model parameters	count
$2P$	Forward-pass FLOPs per token (approximation)	FLOP/token
X_{tokens}	Tokens processed per second per accelerator	tokens/s

From §4.18.6

FORMULA

Batching model

$$T(B) = t_0 + B\,t_1 \qquad X(B) = \frac{B}{t_0 + B\,t_1} \qquad W_{\text{fill}} \approx \frac{B-1}{2\lambda}$$

Symbol	Meaning	Units
B	Batch size	items
t_0	Fixed cost per batch	seconds
t_1	Marginal cost per item	seconds/item
$T(B)$	Time to process one batch	seconds
$X(B)$	Throughput	items/second
λ	Arrival rate of items	items/second
W_{fill}	Average time an item waits for its batch to fill	seconds

Chapter 5 — Computer Architecture

From §5.3.2

FORMULA

CPU execution time (the “iron law”)

$$T_{CPU} = N_{instr} \times CPI \times t_{cycle} = \frac{N_{instr} \times CPI}{f}$$

Symbol	Meaning	Units
T_{CPU}	CPU time to run the program	seconds
N_{instr}	Number of instructions executed	instructions
CPI	Average clock cycles per instruction (its inverse is IPC, instructions per cycle)	cycles/instruction
t_{cycle}	Duration of one clock cycle	seconds/cycle
f	Clock frequency, $1/t_{cycle}$	cycles/second (Hz)

From §5.4.4

FORMULA

Address decomposition in a set-associative cache

$$S = \frac{C}{B \times W}, \qquad \text{offset bits} = \log_2 B, \qquad \text{index bits} = \log_2 S$$

$$\text{set index} = \left\lfloor \frac{A}{B} \right\rfloor \bmod S, \qquad \text{tag} = \left\lfloor \frac{A}{B \times S} \right\rfloor$$

Symbol	Meaning	Units
C	Cache capacity	bytes
B	Line (block) size	bytes
W	Associativity (ways per set)	lines
S	Number of sets	sets
A	Memory address	byte address
tag	High-order address bits stored with the line to identify it	—

From §5.4.5

FORMULA

Average memory access time (AMAT), applied recursively to each level

$$AMAT = t_{L1} + m_{L1} \times (t_{L2} + m_{L2} \times (t_{L3} + m_{L3} \times t_{mem}))$$

Symbol	Meaning	Units
t_{L1}, t_{L2}, t_{L3}	Hit time at each cache level	ns
t_{mem}	DRAM access time	ns
m_{L1}	Fraction of all accesses that miss L1	dimensionless (0–1)
m_{L2}, m_{L3}	**Local** miss rate: fraction of accesses *reaching* that level that miss it	dimensionless (0–1)

From §5.6.1

FORMULA

HDD access time

$$T_{access} = T_{seek} + T_{rot} + T_{transfer}, \qquad T_{rot} = \frac{1}{2} \times \frac{60}{RPM}, \qquad T_{transfer} = \frac{S}{R}$$

$$IOPS_{random} \approx \frac{1}{T_{access}}$$

Symbol	Meaning	Units
T_{seek}	Average time to move the head to the target track	ms
T_{rot}	Average rotational latency: half a revolution	ms
RPM	Spindle speed	revolutions per minute
S	Size of the request	bytes
R	Sustained media transfer rate	bytes/second
$IOPS_{random}$	Random I/O operations per second for one outstanding request	operations/second

From §5.6.3

FORMULA

Write amplification factor and drive lifetime

$$WAF = \frac{\text{bytes written to flash}}{\text{bytes written by the host}}, \qquad \text{Lifetime (days)} \approx \frac{C \times PE}{W_{host} \times WAF}$$

Symbol	Meaning	Units
WAF	Write amplification factor (≥ 1 for a drive without compression)	dimensionless
C	Raw flash capacity	bytes
PE	Rated program/erase cycles per block	cycles
W_{host}	Host write volume per day	bytes/day

From §5.6.5

FORMULA

Little’s Law applied to a storage device

$$IOPS = \frac{QD}{L}, \qquad \text{Throughput} = IOPS \times S$$

Symbol	Meaning	Units
QD	Average number of requests outstanding (queue depth)	requests
L	Average latency per request	seconds
S	Request size	bytes

From §5.8.2

FORMULA

Thread pool sizing heuristic [Goetz et al. 2006, §8.2]

$$N_{threads} = N_{cores} \times U_{target} \times (1 + \frac{W}{C})$$

Symbol	Meaning	Units
$N_{threads}$	Threads in the pool	threads
N_{cores}	Cores available to the pool	cores
U_{target}	Target CPU utilization (0–1)	dimensionless
W	Average time a task spends waiting (I/O, remote calls)	seconds
C	Average CPU time per task	seconds

From §5.9.1

FORMULA

Peak DRAM bandwidth

$$BW_{peak} = R_{transfer} \times W_{bus} \times N_{channels}$$

Symbol	Meaning	Units
$R_{transfer}$	Data transfers per second per channel (the “4800” in DDR5-4800)	transfers/second (MT/s)
W_{bus}	Width of a channel’s data path: 8 bytes (64 bits)	bytes/transfer
$N_{channels}$	Populated memory channels	channels

From §5.12.1

FORMULA

Throughput with batching (fixed plus per-item cost)

$$X(B) = \frac{B}{F + B \times v}, \qquad T_{batch}(B) = F + B \times v, \qquad X_{max} = \frac{1}{v}$$

Symbol	Meaning	Units
B	Items per batch	items
F	Fixed cost per operation (system call, device round trip, interrupt, network RTT)	seconds
v	Variable cost per item	seconds/item
$X(B)$	Throughput at batch size B	items/second
$T_{batch}(B)$	Time to complete one batch	seconds

Chapter 6 — Operating Systems

From §6.4

FORMULA

Context-switch overhead fraction

$$f = \frac{C}{Q + C}$$

Symbol	Meaning	Units
f	Fraction of CPU time lost to switching	dimensionless (0–1)
C	Effective cost of one switch, direct + indirect	µs
Q	Average time a thread runs between switches	µs

From §6.4

FORMULA

Cache refill estimate (indirect cost)

$$C_{\text{refill}} \approx \frac{W}{L} \times \frac{t_{\text{miss}}}{M}$$

Symbol	Meaning	Units
C_{refill}	Time spent reloading the working set after a switch	ns
W	Working set that was evicted	bytes
L	Cache line size (64 on x86-64 and most ARM servers)	bytes/line
t_{miss}	Latency of one miss to the next memory level	ns
M	Memory-level parallelism: misses overlapped in flight	dimensionless (≥ 1)

From §6.5.2

FORMULA

Average turnaround and response time

$$\bar{T}_{\text{turn}} = \frac{1}{n}\sum_{i=1}^{n}\left(T^{i}_{\text{completion}} - T^{i}_{\text{arrival}}\right) \qquad \bar{T}_{\text{resp}} = \frac{1}{n}\sum_{i=1}^{n}\left(T^{i}_{\text{first run}} - T^{i}_{\text{arrival}}\right)$$

Symbol	Meaning	Units
n	Number of jobs	jobs
T^{i}_{arrival}	Time job i became ready	ms
$T^{i}_{\text{first run}}$	Time job i first got a CPU	ms
$T^{i}_{\text{completion}}$	Time job i finished	ms

From §6.5.5

FORMULA

Proportional CPU share

$$s_i = \frac{w_i}{\sum_{j \in R} w_j}$$

Symbol	Meaning	Units
s_i	Fraction of one core's time thread i receives when all threads in R compete	dimensionless (0–1)
w_i	Weight of thread i, from its nice value (nice 0 = 1024; each nice step changes weight by about 1.25×)	dimensionless
R	Set of runnable threads competing on that core	—

From §6.6.3

FORMULA

TLB reach

$$R_{\text{TLB}} = E \times P$$

Symbol	Meaning	Units
R_{TLB}	Amount of memory addressable without a TLB miss	bytes
E	Number of TLB entries	entries
P	Page size per entry	bytes

From §6.6.3

FORMULA

Effective memory access time with page faults

$$\mathrm{EAT} = (1 - p) \times t_{\mathrm{mem}} + p \times t_{\mathrm{fault}}$$

Symbol	Meaning	Units
EAT	Average time per memory access	ns
p	Probability that an access causes a major page fault	dimensionless (0–1)
t_{mem}	Time for an access with no fault	ns
t_{fault}	Time to service a major fault, including the disk read	ns

From §6.14.3

FORMULA

Group-commit throughput

$$X = \frac{B}{t_{\mathrm{fsync}}}$$

Symbol	Meaning	Units
X	Maximum durable commits per second on one log	commits/s
B	Commits made durable by one `fsync` (batch size)	commits
t_{fsync}	Latency of one `fsync`	s

From §6.15.6

FORMULA

Memory for concurrent connections

$$M_{\text{total}} = N \times (m_{\text{unit}} + m_{\text{conn}})$$

Symbol	Meaning	Units
M_{total}	Memory needed for concurrency overhead	bytes
N	Concurrent connections (or in-flight requests)	connections
m_{unit}	Resident memory per concurrency unit (thread stack + kernel task, or coroutine)	bytes
m_{conn}	Per-connection buffers (socket buffers, TLS state, request object)	bytes

From §6.15.6

FORMULA

Thread-pool sizing

$$N_{\text{threads}} = \lambda \times W \qquad \text{and, for mixed CPU/I/O work,} \qquad N_{\text{threads}} \approx N_{\text{cpu}} \times U \times (1 + \frac{T_{\text{wait}}}{T_{\text{cpu}}})$$

Symbol	Meaning	Units
N_{threads}	Threads needed to sustain the load	threads
λ	Request arrival rate	requests/s
W	Time a request occupies a thread	s
N_{cpu}	Available cores	cores
U	Target CPU utilization	dimensionless (0–1)
T_{wait}, T_{cpu}	Per-request time spent waiting and computing	ms

Chapter 7 — Networking Fundamentals

From §7.3.3

FORMULA

Protocol efficiency (goodput fraction)

$$\eta = \frac{P}{P + H}$$

Symbol	Meaning	Units
η	Fraction of transmitted bytes that are application payload	dimensionless (0–1)
P	Application payload bytes per frame	bytes
H	Total per-frame overhead from all layers (headers, trailers, and on Ethernet the preamble and inter-frame gap)	bytes

From §7.4.2

FORMULA

Addresses in a CIDR block

$$N_{\text{addr}} = 2^{W - n} \qquad N_{\text{hosts,IPv4}} = 2^{32 - n} - 2$$

Symbol	Meaning	Units
N_{addr}	Total addresses in the block	addresses
W	Address width: 32 for IPv4, 128 for IPv6	bits
n	Prefix length	bits
$N_{\text{hosts,IPv4}}$	Assignable IPv4 hosts on a traditional subnet: subtract the **network address** (all host bits 0) and the **broadcast address** (all host bits 1)	addresses

From §7.7.5

FORMULA

Retransmission timeout (RFC 6298)

On each new RTT sample R (after the first):

$$\text{RTTVAR} \leftarrow (1-\beta)\,\text{RTTVAR} + \beta\,|\text{SRTT} - R|$$

$$\text{SRTT} \leftarrow (1-\alpha)\,\text{SRTT} + \alpha R$$

$$\text{RTO} = \text{SRTT} + \max(G,\ 4\,\text{RTTVAR})$$

Symbol	Meaning	Units
R	Latest measured round-trip time	ms
SRTT	Smoothed (exponentially averaged) RTT	ms
RTTVAR	Smoothed RTT variation	ms
α, β	Smoothing gains, $1/8$ and $1/4$	dimensionless
G	Clock granularity	ms
RTO	Retransmission timeout; RFC 6298 says round up to at least 1 s	ms

From §7.7.7

FORMULA

CUBIC window growth

$$W(t) = C\,(t-K)^3 + W_{\max}, \qquad K = \sqrt[3]{\frac{W_{\max}\,(1-\beta)}{C}}$$

Symbol	Meaning	Units
$W(t)$	Congestion window t seconds after the last loss	segments
$W_{\max}$	Window just before the last loss	segments
β	Multiplicative decrease factor; window becomes $\beta W_{\max}$ after loss (0.7)	dimensionless
C	Scaling constant (0.4)	segments/s³
K	Time to climb back to $W_{\max}$	s

From §7.15.1

FORMULA

One-hop delay

$$d_{\text{hop}} = \underbrace{\frac{D}{v}}_{\text{propagation}} + \underbrace{\frac{L}{R}}_{\text{transmission}} + d_{\text{queue}} + d_{\text{proc}}$$

Symbol	Meaning	Units
D	Physical length of the link	m (or km)
v	Signal speed in the medium: about 2×10^8 m/s in optical fiber (roughly two-thirds of c)	m/s
L	Packet size	bits
R	Link rate (bandwidth)	bits/s
d_{queue}	Queuing delay: bytes waiting ahead ÷ link rate	s
d_{proc}	Processing delay: microseconds in software, sub-microsecond in switch hardware	s

From §7.15.2

FORMULA

Window-limited throughput and the bandwidth-delay product (BDP)

$$\text{Throughput} \le \frac{W}{\text{RTT}} \qquad \text{BDP} = R \times \text{RTT}$$

Symbol	Meaning	Units
W	Effective window, $\min(\text{cwnd}, \text{rwnd})$	bytes
RTT	Round-trip time	s
R	Bottleneck link rate	bits/s (divide by 8 for bytes/s)
BDP	Data that must be in flight to keep the link full	bits or bytes

From §7.15.3

FORMULA

Mathis throughput formula

$$\text{Throughput} \leq \frac{\text{MSS}}{\text{RTT}} \times \frac{C}{\sqrt{p}}$$

Symbol	Meaning	Units
MSS	Maximum segment size	bits (or bytes, for bytes/s)
RTT	Round-trip time	s
p	Packet loss probability	dimensionless
C	Constant, $\sqrt{3/2} \approx 1.22$ for periodic loss with every packet ACKed	dimensionless

From §7.16.2

FORMULA

Connections needed for a one-request-per-connection protocol

$$N = \lambda \times W$$

Symbol	Meaning	Units
N	Average connections busy	connections
λ	Request rate	requests/s
W	Time a request holds a connection (network RTT plus server time)	s

From §7.17.1

FORMULA

Oversubscription ratio at a leaf

$$\text{Oversubscription} = \frac{\text{total server-facing bandwidth}}{\text{total uplink bandwidth}}$$

Symbol	Meaning	Units
Server-facing bandwidth	Sum of the leaf's downlink port speeds	Gbps
Uplink bandwidth	Sum of the leaf's spine-facing port speeds	Gbps

Chapter 8 — Web Application Fundamentals

From §8.5.1

FORMULA

End-to-end latency of a request on a new connection

$$T_{\text{total}} = T_{\text{DNS}} + T_{\text{TCP}} + T_{\text{TLS}} + T_{\text{req}} + T_{\text{server}} + T_{\text{transfer}}$$

with $T_{\text{TCP}} \approx 1 \times RTT$, $T_{\text{TLS 1.3}} \approx 1 \times RTT$, $T_{\text{req}} \approx 1 \times RTT$ and $T_{\text{transfer}} \approx \frac{S}{B}$ for a response that fits in the congestion window.

Symbol	Meaning	Units
T_{DNS}	DNS lookup time (≈ 0 if cached)	ms
RTT	Client-to-load-balancer round-trip time	ms
T_{server}	Time from the load balancer receiving the request to sending the response (app + DB)	ms
S	Response size	bytes
B	Bottleneck bandwidth	bytes/ms

From §8.6.5

FORMULA

Rows touched per page

$$R_{\text{offset}} = o + \ell \qquad R_{\text{keyset}} \approx \ell + \log_b n$$

Symbol	Meaning	Units
o	Offset (rows skipped)	rows
ℓ	Page size (limit)	rows
n	Rows in the table or index	rows
b	B-tree fan-out (entries per index page, typically hundreds)	entries

From §8.9.2

FORMULA

Varint length

$$\text{bytes}(n) = \max\left(1, \left\lceil \frac{\lfloor \log_2 n \rfloor + 1}{7} \right\rceil\right)$$

Symbol	Meaning	Units
n	Non-negative integer being encoded	—
$\lfloor \log_2 n \rfloor + 1$	Number of significant bits in n (for $n \geq 1$)	bits
7	Payload bits per varint byte	bits/byte

Chapter 9 — Real-Time Communication

From §9.3.1

FORMULA

Short-polling request rate

$$R = \frac{N}{T}$$

Symbol	Meaning	Units
R	Request rate arriving at the servers	requests/second
N	Number of concurrently active (polling) clients	clients
T	Polling interval	seconds

From §9.3.1

FORMULA

Fraction of wasted (empty) polls and added latency

If updates for a client arrive as a Poisson process with rate λ, the probability that a given interval contains no update is

$$P_{\text{empty}} = e^{-\lambda T} \approx 1 - \lambda T \quad (\text{when } \lambda T \ll 1)$$

and the average delay between an event occurring and the client seeing it is

$$D_{\text{avg}} = \frac{T}{2} + t_{\text{rtt}}$$

Symbol	Meaning	Units
P_{empty}	Fraction of polls that return nothing	dimensionless (0–1)
λ	Average update rate per client	events/second
T	Polling interval	seconds
D_{avg}	Average notification delay	seconds
t_{rtt}	Network round-trip time for one poll	seconds

From §9.4.1

FORMULA

Long-polling request rate

With Poisson events at rate λ per client and hold timeout T_h, each request lasts $\min(T_h, X)$ where X is the time to the next event. Its mean is

$$E[\text{duration}] = \frac{1 - e^{-\lambda T_h}}{\lambda}, \qquad R = \frac{N}{E[\text{duration}]}, \qquad C_{\text{open}} \approx N$$

Symbol	Meaning	Units
λ	Average update rate per client	events/second
T_h	Server hold timeout	seconds
$E[\text{duration}]$	Mean lifetime of one long-poll request	seconds
N	Number of connected clients	clients
R	Request rate at the servers	requests/second
C_{open}	Concurrently open requests (Little's Law: $R \times E[\text{duration}]$)	requests

From §9.6.3

FORMULA

WebSocket framing overhead per frame

$$H = 2 + L_{\text{ext}} + 4m, \qquad L_{\text{ext}} = \begin{cases} 0 & n \le 125 \\ 2 & 126 \le n \le 65,535 \\ 8 & n \ge 65,536 \end{cases}$$

Symbol	Meaning	Units
H	Header bytes per frame	bytes
n	Payload length	bytes
L_{ext}	Extended-length bytes	bytes
m	1 if masked (client→server), 0 otherwise	dimensionless

From §9.7.2

FORMULA

Per-participant bandwidth by topology

Topology	Upload per participant	Download per participant	Server work
Mesh	$(N-1)\,b$	$(N-1)\,b$	none
SFU	b (or $\approx 1.5b$–$2b$ with simulcast)	$(N-1)\,b$ (less if receivers get lower layers)	forward $N(N-1)$ streams
MCU	b	b	decode N, mix, encode up to N

Symbol	Meaning	Units
N	Participants in the call	participants
b	Bitrate of one participant's stream	Mbit/s

From §9.9.2

FORMULA

Gateway fleet sizing

$$n_{\text{conn,max}} = \min\left(\frac{f \cdot M}{m},\ c_{\text{blast}}\right) \qquad S = Z \cdot \left\lceil \frac{C}{(Z-1)\,n_{\text{conn,max}}} \right\rceil$$

Symbol	Meaning	Units
M	RAM per gateway server	bytes
f	Fraction of RAM budgeted for connections	dimensionless
m	Memory per connection	bytes/connection
c_{blast}	Maximum connections you accept losing in one server failure	connections
$n_{\text{conn,max}}$	Design ceiling of connections per server	connections
C	Peak concurrent connections	connections
Z	Availability zones; the fleet must survive loss of one	zones
S	Total gateway servers	servers

From §9.9.3

FORMULA

Fan-out load

$$D = \mu \times \bar{r}$$

Symbol	Meaning	Units
D	Message deliveries written to sockets	deliveries/second
μ	Messages published	messages/second
$\bar{r}$	Average number of *online* recipients per message	recipients/message

From §9.9.6

FORMULA

Reconnect arrival rate with uniform jitter

$$\lambda_{\text{reconnect}} \approx \frac{n_{\text{lost}}}{W}$$

Symbol	Meaning	Units
$\lambda_{\text{reconnect}}$	Peak reconnect (handshake) rate	connections/second
n_{lost}	Connections dropped by the failure	connections
W	Width of the random delay window for the first reconnect attempt	seconds

Chapter 10 — Security Fundamentals for System Designers

From §10.3.5

FORMULA

Qualitative risk score

$$R = L \times I$$

Symbol	Meaning	Units
R	Risk score used to rank threats	dimensionless (1–9 on a 3-point scale)
L	Likelihood that the threat is exploited within the planning horizon (1 = unlikely, 2 = plausible, 3 = expected)	ordinal score
I	Impact if exploited (1 = minor, 2 = serious, 3 = severe: data breach, financial loss, regulatory)	ordinal score

From §10.4.1

FORMULA

Birthday bound for hash collisions

$$P_{\text{collision}} \approx \frac{k^2}{2^{n+1}}$$

Symbol	Meaning	Units
k	Number of distinct inputs hashed	count
n	Digest length	bits
$P_{\text{collision}}$	Probability that at least two of the k digests are equal (valid while $P \ll 1$)	probability

Consequence: an n-bit hash offers about $n/2$ bits of collision resistance, because P approaches $1/2$ when $k \approx 2^{n/2}$.

From §10.4.2

FORMULA

Random-nonce collision probability for AES-GCM

$$P_{\text{reuse}} \approx \frac{q^2}{2^{97}}$$

Symbol	Meaning	Units
q	Number of messages encrypted under one key with random 96-bit nonces	messages
P_{reuse}	Probability that any two of them share a nonce (birthday bound with $n = 96$)	probability

From §10.4.7

FORMULA

Entropy of a uniformly random secret

$$H = L \cdot \log_2 N$$

Symbol	Meaning	Units
H	Entropy: the number of bits an attacker must effectively guess	bits
L	Length of the secret	symbols
N	Number of equally likely choices per symbol	symbols

Valid only when every symbol is chosen independently and uniformly at random — true for machine-generated tokens, false for human-chosen passwords.

From §10.6.2

FORMULA

Expected offline brute-force time

$$T = \frac{S}{2 \cdot R \cdot G}$$

Symbol	Meaning	Units
T	Expected time to find one password (average: half the space)	seconds
S	Size of the search space: N^L for random passwords, or the size of the attacker's ranked wordlist up to the target	candidates
R	Guesses per second per attacking device for this hash function and cost	guesses/s/device
G	Number of attacking devices (GPUs)	devices

Salting means T is paid **per user**; without salts, one pass covers all users.

From §10.8

FORMULA

Credential misuse window

$$W = \begin{cases} T_{\text{detect}} + T_{\text{rotate}} & \text{long-lived credential} \\ \min\left(T_{\text{TTL}},\ T_{\text{detect}} + T_{\text{revoke}}\right) & \text{short-lived credential} \end{cases}$$

Symbol	Meaning	Units
W	Time during which a leaked credential remains usable	hours
T_{detect}	Time from leak to detection	hours
$T_{\text{rotate}}, T_{\text{revoke}}$	Time to rotate or revoke after detection	hours
T_{TTL}	Lifetime of a short-lived credential	hours

From §10.9.2

FORMULA

KMS request rate under envelope encryption

$$Q_{\text{KMS}} = \frac{\lambda_{\text{obj}}}{B}$$

Symbol	Meaning	Units
Q_{KMS}	KMS requests generated	requests/s
λ_{obj}	Rate of objects (or records) encrypted or decrypted	objects/s
B	Objects protected per KMS call (1 for a DEK per object; larger with DEK caching or per-batch keys)	objects/request

From §10.11

FORMULA

Amplified attack bandwidth

$$B_{\text{victim}} = B_{\text{attacker}} \times A$$

Symbol	Meaning	Units
B_{victim}	Traffic arriving at the victim	bits/s
B_{attacker}	Spoofed request traffic the attacker sends	bits/s
A	Bandwidth amplification factor of the abused protocol (response bytes / request bytes)	dimensionless

Chapter 11 — Relational Databases

From §11.6.3

FORMULA

I/O cost of join algorithms

$$C_{\text{tuple NLJ}} = M + p_R \cdot M \cdot N \qquad C_{\text{block NLJ}} = M + \lceil \frac{M}{B-2} \rceil N$$

$$C_{\text{index NLJ}} = M + p_R \cdot M \cdot c_{\text{idx}} \qquad C_{\text{Grace hash}} \approx 3(M+N) \qquad C_{\text{sort-merge}} \approx 3(M+N) \text{ to } 5(M+N)$$

Symbol	Meaning	Units
M, N	Pages in the outer relation R and inner relation S	pages
p_R	Rows per page of R	rows/page
B	Buffer pages available to the join (one for output, one for the inner scan)	pages
c_{idx}	Page reads to find matching inner rows through an index (descent plus row fetch)	pages/row
C	Total page reads and writes	page I/Os

From §11.7.5

FORMULA

Break-even selectivity for a secondary-index scan

$$s^* = \frac{P \cdot c_{\text{seq}}}{n \cdot c_{\text{rand}}}$$

Symbol	Meaning	Units
s^*	Fraction of rows matched above which a full scan is cheaper	dimensionless
P	Pages in the table	pages
n	Rows in the table	rows
c_{seq}	Cost of a sequential page read	cost units/page
c_{rand}	Cost of a random page read	cost units/page

From §11.8.2

FORMULA

Height of a B+ tree with L entries per leaf

$$h = 1 + \lceil \log_f \lceil \frac{n}{L} \rceil \rceil, \qquad f = \lfloor \frac{u \cdot P}{e_{\text{int}}} \rfloor, \qquad L = \lfloor \frac{u \cdot P}{e_{\text{leaf}}} \rfloor$$

Symbol	Meaning	Units
h	Levels from root to leaf (page reads per lookup with a cold cache)	levels
n	Entries (rows) indexed	rows
P	Page size	bytes
u	Average fraction of each page holding entries (fill factor minus headers)	dimensionless
e_{int}	Bytes per internal entry (separator key + child pointer + overhead)	bytes
e_{leaf}	Bytes per leaf entry (full row for a clustered index; key + pointer for a secondary index)	bytes
f	Fan-out of internal pages	children/page
L	Entries per leaf page	entries/page

From §11.10.1

FORMULA

Size of the join-order search space

$$N_{\text{left-deep}} = n! \qquad N_{\text{bushy}} = \frac{(2(n-1))!}{(n-1)!} \qquad W_{\text{DP}} \approx n \cdot 2^{n-1}$$

Symbol	Meaning	Units
n	Number of tables joined	tables
$N_{\text{left-deep}}$	Join orders where every join's inner input is a base table	plans
N_{bushy}	All binary join trees, including joins of two intermediate results	plans
W_{DP}	Approximate subproblems examined by Selinger-style dynamic programming over left-deep plans	subproblems

From §11.10.2

FORMULA

Cost of a sequential scan with one filter (PostgreSQL)

$$C_{\text{seq}} = P \cdot c_{\text{seq}} + n \cdot c_{\text{tuple}} + n \cdot k \cdot c_{\text{op}}$$

Symbol	Meaning	Units
P	Heap pages (`pg_class.relpages`)	pages
n	Rows (`pg_class.reltuples`)	rows
k	Operators evaluated per row in the filter	operators
c_{seq} , c_{tuple} , c_{op}	`seq_page_cost` , `cpu_tuple_cost` , `cpu_operator_cost`	cost units

From §11.10.2

FORMULA

Cardinality estimate under the independence assumption

$$\hat{r} = n \cdot \prod_{i=1}^{k} s_i, \qquad s_i \approx \text{freq}(v_i) \text{ or } \frac{1}{d_i}$$

Symbol	Meaning	Units
$\hat{r}$	Estimated rows satisfying all k predicates	rows
n	Rows in the table	rows
s_i	Selectivity of predicate i (fraction of rows passing it)	dimensionless
$\text{freq}(v_i)$	Sampled frequency of value v_i if it is a most-common value	dimensionless
d_i	Number of distinct values in the column	values

From §11.14.2

FORMULA

Time until transaction-ID wraparound

$$T_{\text{wrap}} = \frac{2^{31} - A}{\lambda_{\text{xid}}}$$

Symbol	Meaning	Units
T_{wrap}	Time until PostgreSQL must stop assigning transaction IDs	seconds
A	Current age of the oldest unfrozen transaction ID in the database	transactions
λ_{xid}	Rate of transactions that write (read-only transactions consume no ID)	transactions/second

From §11.15.3

FORMULA

Throughput ceiling of a single hot row

$$X_{\text{max}} = \frac{1}{t_{\text{hold}}}$$

Symbol	Meaning	Units
X_{max}	Maximum transactions per second that can update the same row	transactions/second
t_{hold}	Time each transaction holds the row's X lock (from its update to commit)	seconds

Chapter 12 — NoSQL Databases

From §12.4.5

FORMULA

DynamoDB capacity units

$$\text{WCU} = \sum_{\text{write types}} r_w \times \lceil \frac{s_w}{1\ \text{KB}} \rceil \times t_w \qquad \text{RCU} = \sum_{\text{read types}} r_r \times \lceil \frac{s_r}{4\ \text{KB}} \rceil \times c_r$$

Symbol	Meaning	Units
r_w, r_r	Rate of a given write or read type, including index writes	operations/second
s_w	Size of the item written (for a GSI, the projected size)	KB
s_r	Item size for GetItem; total returned size for Query	KB
t_w	1 for standard writes, 2 for transactional writes	dimensionless
c_r	0.5 eventually consistent, 1 strongly consistent, 2 transactional	dimensionless

From §12.4.5

FORMULA

On-demand versus provisioned break-even utilization

$$u^{*} = \frac{p_{\text{prov}}/3600}{p_{\text{od}}}$$

Symbol	Meaning	Units
u^{*}	Average utilization of provisioned capacity above which provisioned is cheaper	fraction (0–1)
p_{prov}	Price of one provisioned capacity unit for one hour	USD per unit-hour
p_{od}	Price of one on-demand request unit	USD per request unit

From §12.6.4

FORMULA

Cassandra partition cells and size

$$N_v = N_r \, (N_c - N_{pk} - N_s) + N_s$$

$$S_t = \sum_i \text{sizeOf}(c_{k_i}) + \sum_j \text{sizeOf}(c_{s_j}) + N_r \times \left(\sum_k \text{sizeOf}(c_{r_k}) + \sum_l \text{sizeOf}(c_{c_l}) \right) + N_v \times \text{sizeOf}(t_{\text{avg}})$$

Symbol	Meaning	Units
N_v	Number of cells (values) in the partition	cells
N_r	Number of rows in the partition	rows
N_c	Number of columns in the table	columns
N_{pk}	Number of partition-key columns	columns
N_s	Number of static columns	columns
S_t	Estimated partition size on disk, one replica, before compression	bytes
c_k, c_s, c_r, c_c	Partition-key, static, regular and clustering columns	bytes each
t_{avg}	Average per-cell metadata (the docs use the 8-byte timestamp)	bytes

From §12.7.3

FORMULA

Cost of a k-hop traversal

$$E_k = \sum_{i=1}^{k} d^i \qquad T_{\text{graph}} \approx E_k \times t_{\text{ptr}} \qquad T_{\text{join}} \approx E_k \times t_{\text{idx}}$$

Symbol	Meaning	Units
E_k	Edges expanded by a k-hop traversal without pruning	edges
d	Average out-degree (neighbors per node)	edges/node
k	Number of hops	hops
t_{ptr}	Cost of following one adjacency pointer	µs
t_{idx}	Cost of one index lookup on the edge table, growing with $\log N$	µs

From §12.8.3

FORMULA

Tiered time-series storage

$$S = N_{\text{series}} \times \sum_{i} \frac{R_i}{\Delta_i} \times b_i$$

Symbol	Meaning	Units
S	Total storage	bytes
N_{series}	Number of active series	series
R_i	Retention period of tier i	seconds
Δ_i	Sample (or rollup) interval of tier i	seconds
b_i	Bytes per stored point in tier i after compression	bytes/point

Chapter 13 — Storage Engine Internals

From §13.3.2

FORMULA

Records per slotted page

$$n = \left\lfloor \frac{P - H}{r + s} \right\rfloor$$

Symbol	Meaning	Units
n	Records that fit in one page	records
P	Page size	bytes
H	Page header size	bytes
r	Stored record size including per-record header and alignment padding	bytes
s	Slot (line pointer) size	bytes

From §13.4.3

FORMULA

Break-even reference interval (the “five-minute rule”)

$$T_{BE} = \frac{C_{dev} / IOPS_{dev}}{C_{RAM} \times P}$$

Symbol	Meaning	Units
T_{BE}	Break-even interval: cache a page if it is re-referenced more often than once per T_{BE}	seconds
C_{dev}	Price of one storage device	dollars
$IOPS_{dev}$	Random reads per second the device sustains	reads/s
C_{RAM}	Price of memory	dollars/byte
P	Page size	bytes

From §13.5.3

FORMULA

Redo work after a crash

$$W_{redo} \approx r_{WAL} \times \Delta t_{ckpt}, \qquad T_{redo} \approx \frac{W_{redo}}{b_{replay}}$$

Symbol	Meaning	Units
W_{redo}	Log volume between the redo start point and the crash	bytes
r_{WAL}	Rate at which the workload generates log	bytes/s
Δt_{ckpt}	Time between checkpoints (worst case, crash just before the next one)	s
T_{redo}	Duration of the redo pass	s
b_{replay}	Replay throughput, usually limited by random page reads, not log reads	bytes/s

From §13.8.1

FORMULA

Number of levels in a leveled LSM tree

$$L = \lceil \log_T \frac{N}{S_1} \rceil + 1 \quad (\text{levels } L_1 \ldots L_L, \text{ for } N > S_1)$$

Symbol	Meaning	Units
L	Number of levels below L0	levels
N	Total data size	bytes
S_1	Target size of level 1	bytes
T	Size ratio (fan-out) between adjacent levels	dimensionless

From §13.9.1

FORMULA

Approximate amplification of compaction strategies

$$WA_{level} \approx 2 + L\,\frac{T+1}{2}, \qquad WA_{tier} \approx 2 + L, \qquad WA_{lazy} \approx 2 + (L-1) + \frac{T+1}{2}$$

$$R_{level} \approx L, \qquad R_{tier} \approx (T-1)L, \qquad R_{lazy} \approx (T-1)(L-1) + 1$$

Symbol	Meaning	Units
WA	Write amplification: bytes written to storage per byte written by the application (the 2 counts the WAL and the memtable flush)	dimensionless
R	Sorted runs a point lookup may probe in the worst case (excluding memtables)	runs
L	Number of levels (tiers), $\lceil \log_T (N/M) \rceil$	levels
T	Size ratio between levels; runs merged at once in tiering	dimensionless
N	Data size	bytes
M	Memtable flush size	bytes

From §13.10

FORMULA

Bloom filter false-positive rate (summary of §17.4)

$$p \approx (1 - e^{-kn/m})^k, \qquad k_{opt} = \frac{m}{n} \ln 2, \qquad p_{opt} \approx 0.6185^{m/n}$$

Symbol	Meaning	Units
p	Probability that an absent key gets “probably yes”	dimensionless
m	Bits in the filter	bits
n	Keys inserted	keys
k	Hash functions	count

From §13.11.1

FORMULA

The three amplifications

$$RA = \frac{\text{bytes read from storage}}{\text{bytes returned}}, \qquad WA = \frac{\text{bytes written to storage}}{\text{bytes written by the application}}, \qquad SA = \frac{\text{bytes on storage}}{\text{bytes of live data}}$$

Symbol	Meaning	Units
RA	Read amplification (often counted instead as I/Os per lookup)	dimensionless
WA	Write amplification at the engine level, before the SSD’s own WAF	dimensionless
SA	Space amplification	dimensionless

From §13.13.2

FORMULA

Size of a dictionary-encoded, bit-packed column

$$S \approx \frac{N\lceil \log_2 d \rceil}{8} + d \cdot \bar{v}$$

Symbol	Meaning	Units
S	Encoded column size	bytes
N	Number of rows	rows
d	Number of distinct values	values
$\bar{v}$	Average size of one dictionary entry	bytes

Chapter 14 — Database Scaling: Replication, Partitioning and Sharding

From §14.3.1

FORMULA

Read capacity of a replicated group

$$R_{\max} = n \cdot (u \cdot c - w) \qquad \text{valid only while } w < u \cdot c$$

Symbol	Meaning	Units
$R_{\max}$	Maximum sustainable read rate across the group	reads/s
n	Number of nodes serving reads (replicas, plus the primary if it serves reads)	nodes
c	Capacity of one node, expressed in read-equivalent operations	ops/s
u	Target utilization, leaving head-room for spikes and failover	dimensionless (0–1)
w	Cost of applying the full write stream on one node, in read-equivalent operations	ops/s

From §14.4.1

FORMULA

Commit latency with k acknowledgments

$$T_{\text{commit}} \approx T_{\text{local}} + \text{RTT}_{(k)} + T_{\text{remote}}$$

Symbol	Meaning	Units
T_{commit}	Time from the client's `COMMIT` to the primary's reply	ms
T_{local}	Time to flush the log on the primary	ms
$\text{RTT}_{(k)}$	The k-th smallest round-trip time among the eligible replicas	ms
T_{remote}	Work the replica does before acknowledging (write, flush or apply)	ms
k	Number of replica acknowledgments required ($k = 0$ for asynchronous)	count

From §14.4.2

FORMULA

Replica catch-up time

$$T_{\text{catch-up}} = \frac{B}{r_{\text{apply}} - r_{\text{in}}} \qquad (r_{\text{apply}} > r_{\text{in}})$$

Symbol	Meaning	Units
$T_{\text{catch-up}}$	Time until the replica's lag returns to near zero	s
B	Backlog of unapplied changes when catch-up starts	transactions
r_{apply}	Rate at which the replica can apply changes	transactions/s
r_{in}	Rate at which the primary generates new changes during catch-up	transactions/s

From §14.7.4

FORMULA

Number of shards required

$$S = \lceil \max\left(\frac{D}{C_{\text{disk}} \cdot f},\ \frac{W}{w_{\text{node}} \cdot u},\ \frac{R}{n_{\text{read}} \cdot r_{\text{node}} \cdot u}\right) \rceil$$

Symbol	Meaning	Units
S	Number of shards (each one a replicated group)	shards
D	Data size at the planning horizon, one copy, including indexes	TB
C_{disk}	Usable storage per node	TB
f	Target fill fraction, leaving room for compaction, backups and growth between reshards	dimensionless (0–1)
W	Peak write rate at the horizon	writes/s
w_{node}	Sustainable write rate of one primary (from benchmarks)	writes/s
R	Peak read rate that reaches the database, after caches	reads/s
n_{read}	Nodes per shard that serve reads	nodes
r_{node}	Sustainable read rate of one node	reads/s
u	Target utilization at peak	dimensionless (0–1)

From §14.8.3

FORMULA

Expected keys moved on a membership change

$$E[\text{moved on add}] = \frac{K}{N+1} \qquad E[\text{moved on remove}] = \frac{K}{N}$$

Symbol	Meaning	Units
K	Total number of keys	keys
N	Number of nodes before the change	nodes

From §14.8.4

FORMULA

Load imbalance on a ring with virtual nodes

$$\mathrm{CV}(N,V) = \frac{\sqrt{\mathrm{Var}[L]}}{E[L]} = \sqrt{N-1}\,\sqrt{NV+1} \;\approx\; \frac{1}{\sqrt{V}} \quad (N \gg 1)$$

$$\frac{E[\max_i L_i]}{1/N} \approx 1 + z_N \cdot \mathrm{CV}(N,V) \quad (V \gtrsim 16), \qquad \frac{E[\max_i L_i]}{1/N} = H_N \quad (V = 1)$$

Symbol	Meaning	Units
L, L_i	Fraction of the key space owned by a node (by node i)	dimensionless (0–1)
N	Number of physical nodes	nodes
V	Virtual nodes per physical node	vnodes/node
CV	Coefficient of variation: standard deviation of a node's load divided by its mean	dimensionless
z_N	Expected maximum of N independent standard normal variables: 1.16 for $N = 5$, 1.54 for $N = 10$, 2.51 for $N = 100$, 3.24 for $N = 1,000$	dimensionless
H_N	N-th harmonic number $1 + \frac{1}{2} + \ldots + \frac{1}{N} \approx \ln N + 0.577$	dimensionless

From §14.10.1

FORMULA

Salt factor for a hot key

$$s = \left\lceil \frac{\lambda_{\text{key}}}{u \cdot c_{\text{part}}} \right\rceil \qquad \text{read amplification} = s$$

Symbol	Meaning	Units
s	Number of sub-keys (salt factor)	sub-keys
λ_{key}	Write rate to the hot key	writes/s
c_{part}	Write capacity of one partition	writes/s
u	Target utilization of a partition	dimensionless (0–1)

From §14.13.1

FORMULA

Server connections needed behind a transaction pooler (Little's Law)

$$C_{\text{server}} = \lambda_{\text{tx}} \cdot W_{\text{tx}} \cdot (1 + h)$$

Symbol	Meaning	Units
C_{server}	Server connections the pool needs	connections
λ_{tx}	Transaction rate through the pooler	transactions/s
W_{tx}	Time a transaction holds a server connection, from `BEGIN` to `COMMIT`	s
h	Headroom fraction for bursts	dimensionless

Chapter 15 — Object, Block and File Storage

From §15.3.1

FORMULA

Access time for one read

$$t \approx t_{\text{first}} + \frac{S}{B} \qquad t_{\text{parallel}} \approx t_{\text{first}} + \frac{S}{\min\left(c \cdot B,\ B_{\text{NIC}}\right)}$$

Symbol	Meaning	Units
t	Time to read the whole request	s
t_{first}	Time to first byte (network round trip, lookup, seek or flash read)	s
S	Bytes requested	bytes
B	Throughput of one stream (connection or volume)	bytes/s
c	Number of parallel streams (for example byte-range GETs)	count
B_{NIC}	Client network interface limit	bytes/s

From §15.4.2

FORMULA

Prefixes needed for a target request rate

$$N_{\text{prefix}} = \lceil \frac{R_{\text{peak}}}{r_{\text{prefix}}} \rceil$$

Symbol	Meaning	Units
N_{prefix}	Distinct key prefixes over which requests must be spread	count
R_{peak}	Peak request rate of one type (writes or reads)	requests/s
r_{prefix}	Sustained rate one partitioned prefix supports	requests/s

From §15.5.1

FORMULA

Metadata memory for an in-memory namespace

$$M = N_{\text{files}} \cdot b_{\text{file}} + N_{\text{chunks}} \cdot b_{\text{chunk}} \qquad N_{\text{chunks}} \approx \sum_{\text{files}} \lceil \frac{\text{file size}}{C} \rceil$$

Symbol	Meaning	Units
M	Memory needed by the metadata server	bytes
N_{files}	Number of files (and directories)	count
N_{chunks}	Number of chunks or blocks	count
$b_{\text{file}}, b_{\text{chunk}}$	Bytes of metadata per file and per chunk	bytes
C	Chunk (block) size	bytes

From §15.6.1

FORMULA

Overhead, fault tolerance and repair reads of a (k, m) code

$$\text{overhead} = \frac{k+m}{k} \qquad \text{tolerated losses} = m \qquad \text{repair read per lost fragment} = r \times F$$

Symbol	Meaning	Units
k	Data fragments per stripe	count
m	Parity fragments per stripe	count
overhead	Raw bytes stored per logical byte	dimensionless (×)
r	Fragments read to rebuild one: 1 for replication, k for RS, local group size for LRC	count
F	Fragment size	bytes

From §15.6.1

FORMULA

Approximate annual probability of losing a stripe (generalizing §4.16)

$$P_{\text{loss/yr}} \approx \frac{n!}{(n-m-1)!}\,\lambda^{m+1}\,T^{m} \times H, \qquad n = k+m, \quad \lambda T \ll 1$$

Symbol	Meaning	Units
n	Fragments per stripe (replicas for replication)	count
m	Losses tolerated	count
λ	Per-fragment (per-disk) failure rate, AFR/8,760	failures/hour
T	Mean time to detect and rebuild one lost fragment	hours
H	Hours per year (8,760)	hours/year

From §15.6.1

FORMULA

Daily repair read traffic in a fleet

$$R_{\text{day}} = \frac{N_{\text{disk}} \times \text{AFR}}{365} \times D_{\text{used}} \times r$$

Symbol	Meaning	Units
R_{day}	Bytes read per day to rebuild failed disks	bytes/day
N_{disk}	Disks in the fleet	count
AFR	Annualized failure rate	fraction/year
D_{used}	Data stored per disk	bytes
r	Fragments read per fragment rebuilt	count

From §15.8.1

FORMULA

Part sizing and upload time

$$p_{\min} = \max\left(5\ \text{MiB}, \frac{S}{N_{\max}}\right) \qquad N = \left\lceil \frac{S}{p} \right\rceil \qquad t \approx \frac{S}{\min\left(c \cdot b,\ B_{\text{NIC}}\right)}$$

Symbol	Meaning	Units
S	Object size	bytes
p, $p_{\min}$	Chosen and minimum part size	bytes
$N_{\max}$	Maximum parts per upload (10,000 in S3)	count
N	Number of parts	count
c	Concurrent part uploads	count
b	Throughput per connection	bytes/s
B_{NIC}	Client network limit	bytes/s

From §15.8.3

FORMULA

Expected chunk size and deduplication index size

$$E[\text{chunk}] \approx s_{\min} + 2^{b} \qquad M_{\text{index}} = \frac{U}{E[\text{chunk}]} \times e$$

Symbol	Meaning	Units
b	Number of hash bits that must match (boundary probability 2^{-b} per byte)	bits
$s_{\min}$	Minimum chunk size (no boundary is accepted earlier)	bytes
$E[\text{chunk}]$	Expected chunk size (ignoring the maximum cap)	bytes
U	Unique data stored	bytes
e	Index entry size (hash plus location)	bytes

From §15.9

FORMULA

Cost of one cohort of data over its lifetime

$$C = \sum_{\text{tiers } i} V \cdot p_i \cdot t_i \; + \sum_{\text{transitions } j} N_{\text{obj}} \cdot q_j \; + V_{\text{read}} \cdot g$$

Symbol	Meaning	Units
V	Volume of the cohort	GB
p_i, t_i	Storage price and months spent in tier i	$/GB-month, months
N_{obj}	Number of objects	count
q_j	Price per lifecycle transition request into tier j	$/request
V_{read}, g	Data retrieved and its per-GB retrieval fee	GB, $/GB

Chapter 16 — Search Systems

From §16.3.4

FORMULA

Variable-byte length of an integer

$$\text{bytes}(g) = \max\left(1, \left\lceil \frac{\lfloor \log_2 g \rfloor + 1}{7} \right\rceil\right)$$

Symbol	Meaning	Units
g	The integer to encode, here a docID gap ($g \geq 1$)	count
$\lfloor \log_2 g \rfloor + 1$	Number of significant bits in g	bits
$\text{bytes}(g)$	Encoded length, 7 payload bits per byte	bytes

From §16.3.5

FORMULA

Inverted index size (one copy)

$$S_{\text{index}} \approx \underbrace{N \cdot u \cdot (b_d + b_f)}_{\text{docIDs + frequencies}} + \underbrace{N \cdot L \cdot b_p}_{\text{positions}} + \underbrace{M \cdot b_t}_{\text{dictionary}}$$

Symbol	Meaning	Units
N	Number of documents	documents
u	Average number of *distinct* terms per document (postings per document)	postings/document
L	Average number of tokens per document (one position each)	tokens/document
b_d, b_f	Compressed bytes per docID gap and per term frequency	bytes
b_p	Compressed bytes per position gap	bytes
M	Number of distinct terms in the collection	terms
b_t	Dictionary bytes per term (term text + pointers + statistics)	bytes

From §16.5.2

FORMULA

TF-IDF weight of term t in document d

$$w_{t,d} = \mathrm{tf}_{t,d} \times \log_{10} \frac{N}{\mathrm{df}_t} \qquad \mathrm{score}(q,d) = \sum_{t \in q} w_{t,d}$$

Symbol	Meaning	Units
$\mathrm{tf}_{t,d}$	Occurrences of term t in document d	count
N	Documents in the collection	documents
df_t	Documents containing t	documents
$w_{t,d}$	Weight of t in d	dimensionless

From §16.5.3

FORMULA

BM25 inverse document frequency (Lucene form)

$$\mathrm{IDF}(t) = \ln\left(1 + \frac{N - n_t + 0.5}{n_t + 0.5}\right)$$

Symbol	Meaning	Units
N	Documents in the collection (or shard; see §16.7.5)	documents
n_t	Documents containing term t (document frequency)	documents
$\mathrm{IDF}(t)$	Weight of a match on t	dimensionless

From §16.5.3

FORMULA

BM25 score

$$\text{score}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f_{t,d}\,(k_1 + 1)}{f_{t,d} + k_1\left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}$$

Symbol	Meaning	Units
$f_{t,d}$	Frequency of term t in document (field) d	count
$\lvert d\rvert$	Length of d in terms after analysis	terms
avgdl	Average length of that field across the collection	terms
k_1	Term-frequency saturation parameter; Lucene default 1.2 (typical 1.2–2.0)	dimensionless
b	Length-normalization strength, 0 (none) to 1 (full); default 0.75	dimensionless
$\text{score}(q,d)$	Relevance score; only comparable within one query	dimensionless

From §16.6.3

FORMULA

Write amplification of tiered merging

$$W \approx 1 + \log_m \frac{S_{\text{max}}}{S_{\text{flush}}}$$

Symbol	Meaning	Units
W	Bytes written to disk per byte of new index data	dimensionless
m	Merge factor: segments merged together per tier	count
S_{max}	Maximum merged segment size	bytes
S_{flush}	Typical size of a newly flushed segment	bytes

From §16.7.3

FORMULA

Scatter-gather tail latency

$$P(\text{query fast}) = p^S \qquad q_{\text{shard}} = q_{\text{query}}^{1/S}$$

Symbol	Meaning	Units
p	Probability one shard answers within threshold T	dimensionless
S	Number of shards the query waits for	count
$P(\text{query fast})$	Probability all S shards answer within T	dimensionless
q_{query}	Target percentile for the whole query (e.g. 0.99)	dimensionless
q_{shard}	Percentile each shard must meet at T	dimensionless

From §16.7.4

FORMULA

Deep pagination cost at the coordinator

$$E = S \cdot (\mathit{from} + \mathit{size})$$

Symbol	Meaning	Units
S	Shards queried	count
from	Offset of the first requested result	results
size	Results per page	results
E	Entries every query phase must return and the coordinator must merge	entries

From §16.9.1

FORMULA

Autocomplete load and precomputed-index memory

$$\lambda_{ac} = \lambda_{search} \cdot c \cdot f \qquad M_{top\text{-}k} \approx P \cdot k \cdot b$$

Symbol	Meaning	Units
λ_{search}	Searches started per second	searches/s
c	Characters typed per search before selecting or submitting	characters
f	Fraction of keystrokes that send a request (after debounce and client caching)	dimensionless
λ_{ac}	Suggestion request rate	requests/s
P	Number of trie nodes (distinct prefixes) with stored suggestions	nodes
k	Suggestions stored per prefix	count
b	Bytes per stored suggestion reference	bytes
$M_{top\text{-}k}$	Memory for the precomputed lists	bytes

From §16.10.3

FORMULA

End-to-end freshness latency

$$T_{\text{fresh}} \approx T_{\text{cdc}} + T_{\text{queue}} + T_{\text{batch}} + T_{\text{index}} + T_{\text{refresh}}$$

Symbol	Meaning	Units
T_{cdc}	Commit to event published in Kafka	s
T_{queue}	Time the event waits for the indexer (consumer lag)	s
T_{batch}	Wait for a bulk request to fill or its timer to fire	s
T_{index}	Bulk request execution on primary and replicas	s
T_{refresh}	Wait for the next refresh: uniform in $[0, I]$, mean $I/2$	s
T_{fresh}	Commit in source database to searchable	s

Chapter 17 — Data Structures and Algorithms for Scale

From §17.3.3

FORMULA

Expected bucket load and the chi-square uniformity test

$$\mu = \frac{n}{B}, \qquad \sigma \approx \sqrt{\frac{n}{B}}, \qquad \chi^2 = \sum_{i=1}^{B} \frac{(c_i - \mu)^2}{\mu}, \qquad \mathbb{E}[\chi^2] \approx B - 1, \quad \text{sd}(\chi^2) \approx \sqrt{2(B-1)}$$

Symbol	Meaning	Units
n	Number of keys hashed	keys
B	Number of buckets	buckets
μ	Expected keys per bucket	keys/bucket
σ	Standard deviation of a bucket's count under a uniform hash	keys/bucket
c_i	Observed number of keys in bucket i	keys
χ^2	Chi-square statistic: total squared deviation from uniform	dimensionless

From §17.4.3

FORMULA

Bloom filter false-positive rate

$$p \approx \left(1 - e^{-kn/m}\right)^{k}$$

Symbol	Meaning	Units
p	Probability that a query for an item not in the set returns “probably present”	probability (0–1)
n	Number of items inserted	items
m	Size of the bit array	bits
k	Number of hash functions (bit positions per item)	count
$e^{-kn/m}$	Approximate fraction of bits still 0	fraction

From §17.4.3

FORMULA

Optimal number of hash functions

$$k^{*} = \frac{m}{n} \ln 2 \approx 0.693 \frac{m}{n}, \qquad p\big|_{k = k^{*}} = \left(\tfrac{1}{2}\right)^{k^{*}}$$

Symbol	Meaning	Units
k^{*}	Number of hash functions that minimizes p (round to an integer)	count
m/n	Bits per element	bits/item
$\ln 2$	Natural logarithm of 2, ≈ 0.693	dimensionless

From §17.4.3

FORMULA

Bits per element for a target false-positive rate

$$\frac{m}{n} = \frac{-\ln p}{(\ln 2)^2} \approx 1.44 \log_2 \frac{1}{p}, \qquad k = -\log_2 p$$

Symbol	Meaning	Units
m/n	Bits of filter needed per inserted item	bits/item
p	Target false-positive rate	probability
$(\ln 2)^2$	≈ 0.4805	dimensionless
k	Matching number of hash functions (rounded)	count

From §17.4.7

FORMULA

Cuckoo filter false-positive bound and space cost

$$\varepsilon \le \frac{2b}{2^f}, \qquad f \ge \log_2 \frac{2b}{\varepsilon}, \qquad \text{bits per item} = \frac{f}{\alpha} \approx \frac{\log_2 (1/\varepsilon) + \log_2 (2b)}{\alpha}$$

Symbol	Meaning	Units
ε	False-positive rate	probability
b	Slots per bucket (4 is the common choice)	slots
f	Fingerprint length	bits
α	Load factor reached before inserts fail (≈ 0.95 for $b = 4$)	fraction

From §17.5.1

FORMULA

HyperLogLog estimate and standard error

$$E = \alpha_m m^2 \left(\sum_{j=1}^{m} 2^{-M[j]} \right)^{-1}, \qquad \alpha_m \approx \frac{0.7213}{1 + 1.079/m}, \qquad \text{SE} \approx \frac{1.04}{\sqrt{m}}, \qquad \text{memory} = m \times 6 \text{ bits}$$

Symbol	Meaning	Units
E	Estimated number of distinct items	items
m	Number of registers, $m = 2^p$	registers
$M[j]$	Value of register j: the largest first-1-bit position seen	count
α_m	Bias-correction constant (valid for $m \geq 128$)	dimensionless
SE	Relative standard error of E	fraction
6 bits	Register width: enough to hold values up to 63, sufficient for a 64-bit hash	bits

From §17.6.1

FORMULA

Count-Min sizing and error bound

$$w = \lceil \frac{e}{\varepsilon} \rceil, \qquad d = \lceil \ln \frac{1}{\delta} \rceil, \qquad f_x \leq \hat{f}_x \leq f_x + \varepsilon N \text{ with probability } \geq 1 - \delta$$

Symbol	Meaning	Units
w	Counters per row (width)	counters
d	Rows, one hash function each (depth)	rows
ε	Error as a fraction of the stream total	fraction
δ	Probability that the error bound is exceeded for a given query	probability
N	Sum of all updates in the stream	events
f_x, $\hat{f}_x$	True and estimated count of item x	events
e	Euler's number, ≈ 2.718	dimensionless

From §17.6.2

FORMULA

Space-Saving guarantee

$$c_x - f_x \le c_{\min} \le \frac{N}{k}, \qquad f_x > \frac{N}{k} \Rightarrow x \text{ is monitored}$$

Symbol	Meaning	Units
k	Number of counters	counters
N	Stream length	events
c_x, f_x	Reported and true count of monitored item x	events
$c_{\min}$	Smallest counter value	events

From §17.7.1

FORMULA

Skip list levels, search cost and memory

$$L(n) = \log_{1/p} n, \qquad \mathbb{E}[\text{search path}] \le \frac{L(n)}{p} + \frac{1}{1-p}, \qquad \mathbb{E}[\text{pointers per node}] = \frac{1}{1-p}$$

Symbol	Meaning	Units
n	Number of elements	elements
p	Probability of promoting a node to the next level	probability
$L(n)$	Expected number of levels in use	levels
Search path	Forward moves plus level drops during one search	steps
Pointers per node	Average forward pointers stored per element	pointers

From §17.8.1

FORMULA

Merkle proof size and anti-entropy cost

$$h = \lceil \log_2 n \rceil, \qquad \text{proof size} = h \times H_{\text{bytes}}, \qquad \text{hashes compared} \le 1 + 2\,k\,h$$

Symbol	Meaning	Units
n	Number of leaves (records or key ranges)	leaves
h	Tree height	levels
H_{bytes}	Size of one hash (32 for SHA-256)	bytes
k	Number of divergent leaves between two replicas	leaves

From §17.9.2

FORMULA

Geohash cell size

$$b_{\text{lon}} = \lceil \frac{5L}{2} \rceil, \quad b_{\text{lat}} = \lfloor \frac{5L}{2} \rfloor, \quad W = \frac{360^\circ}{2^{b_{\text{lon}}}} \times 111.32 \cos\varphi \text{ km}, \quad H = \frac{180^\circ}{2^{b_{\text{lat}}}} \times 110.57 \text{ km}$$

Symbol	Meaning	Units
L	Geohash length	characters
b_{lon}, b_{lat}	Bits spent on longitude and latitude	bits
W, H	Cell width (east–west) and height (north–south)	km
φ	Latitude of the cell	degrees
111.32, 110.57	Approximate km per degree of longitude at the equator and of latitude	km/degree

From §17.9.4

FORMULA

Average S2 cell area at level L

$$\bar{A}_L \approx \frac{4\pi R^2}{6 \times 4^L} = \frac{510.1 \times 10^6 \text{ km}^2}{6 \times 4^L}$$

Symbol	Meaning	Units
$\bar{A}_L$	Average area of a level-L cell	km²
R	Earth's mean radius, ≈ 6,371 km	km
L	Cell level, 0 (a cube face) to 30	level

From §17.10.2

FORMULA

Birthday bound for random identifiers

$$P_{\text{collision}} \approx 1 - e^{-n^2/(2N)} \approx \frac{n^2}{2N} \text{ (when small)}, \qquad n_{50\%} \approx \sqrt{2N \ln 2} \approx 1.18\sqrt{N}$$

Symbol	Meaning	Units
n	Number of IDs generated within one collision domain	IDs
N	Number of possible values, $2^{\text{random bits}}$	values
$P_{\text{collision}}$	Probability that at least two IDs are equal	probability
$n_{50\%}$	Number of IDs at which a collision becomes a coin flip	IDs

From §17.10.3

FORMULA

Snowflake capacity and lifetime

$$\text{rate}_{\text{machine}} = 2^{S} \times 1000 \text{ IDs/s}, \qquad \text{rate}_{\text{cluster}} = 2^{M} \times \text{rate}_{\text{machine}}, \qquad \text{lifetime} = \frac{2^{T}}{1000 \times 3600 \times 24 \times 365.25} \text{ yea}$$

Symbol	Meaning	Units
T	Timestamp bits (millisecond resolution)	bits
M	Machine-ID bits	bits
S	Sequence bits	bits
rate	Maximum sustained ID generation rate	IDs/s
lifetime	Time from the epoch until the timestamp field overflows	years

From §17.12.1

FORMULA

Reservoir sampling inclusion probability

$$P(\text{item } i \text{ in sample after } n \text{ items}) = \frac{k}{i} \times \prod_{j=i+1}^{n} \left(1 - \frac{1}{j}\right) = \frac{k}{i} \times \frac{i}{n} = \frac{k}{n}, \qquad \mathbb{E}[\text{replacements}] \approx k \ln \frac{n}{k}$$

Symbol	Meaning	Units
k	Reservoir size	items
n	Items seen so far	items
i	Arrival position of a particular item ($i > k$)	position
$1 - 1/j$	Probability that item j does not evict this item: it is chosen with probability k/j and then evicts one of k slots	probability